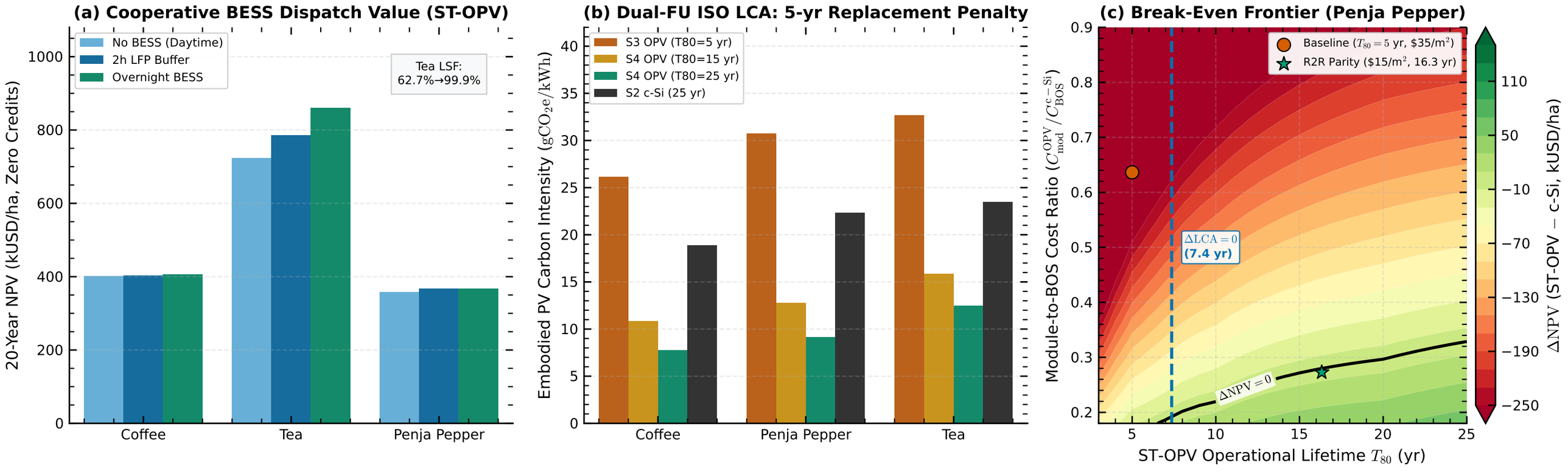

(a) Cooperative BESS Dispatch Value (ST-OPV)
No BESS (Daytime)
2h LFP Buffer
Overnight BESS
Tea LSF:
62.7%→99.9%
20-Year NPV (kUSD/ha, Zero Credits)
0
200
400
600
800
1000
Coffee
Tea
Penja Pepper
(b) Dual-FU ISO LCA: 5-yr Replacement Penalty
S3 OPV (T80=5 yr)
S4 OPV (T80=15 yr)
S4 OPV (T80=25 yr)
S2 c-Si (25 yr)
Embodied PV Carbon Intensity (gCO2e/kWh)
0
5
10
15
20
25
30
35
40
Coffee
Penja Pepper
Tea
(c) Break-Even Frontier (Penja Pepper)
Baseline (T80 = 5 yr, $35/m2)
R2R Parity ($15/m2, 16.3 yr)
ΔLCA = 0
(7.4 yr)
ΔNPV = 0
Module-to-BOS Cost Ratio (C_mod^OPV / C_BOS^c−Si)
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
5
10
15
20
25
ST-OPV Operational Lifetime T80 (yr)
ΔNPV (ST-OPV − c-Si, kUSD/ha)
110
50
−10
−70
−130
−190
−250